



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (Credit System) Degree Examinations

Supplementary Examinations - December 2025

MATDIP401 – ADVANCED MATHEMATICS – II (FOR LATERAL ENTRY STUDENTS ONLY)

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from Unit – I and Two full questions from Unit – II & Unit – III each.

Unit – I		Marks	BT*	CO*	PO*
1. a)	For any two events A and B , prove that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.	6	L*3	1	1
b)	The probability that a communication system will have high fidelity is 0.81. The probability that the system will have both high fidelity and high selectivity is 0.18. Find the probability that system will have high selectivity given that it has high fidelity.	7	L2	1	2
c)	State and prove Bayes' theorem.	7	L2	1	2
2. a)	A class contains 10 men and 20 women of which half the men and half the women have brown eyes. Find the probability that a person chosen at random is a man or has brown eyes?	6	L2	1	2
b)	State Kolmogorov's axioms of probability. Prove that i) $P(A) = 1 - P(\bar{A})$ ii) $P(\phi) = 0$.	7	L3	1	1
c)	A committee consists of 9 students, 2 of which are from first year, 3 are from second year and 4 are from third year. Three students are selected at random. What is the probability that i) All belong to the same class. ii) Two belong to the same class and third belong to the different class. iii) All belong to different classes.	7	L2	1	2
Unit – II					
3. a)	Define order and degree of a differential equation. Find order and degree of the following differential equations: (i) $\left(\frac{d^3x}{dt^3}\right)^2 - t^2 \frac{dx}{dt} + x = 0$ (ii) $27 \frac{d^2y}{dx^2} + 35 \frac{dy}{dx} + \frac{1}{22}y = 75 \cos 2x$	6	L3	2	2
b)	Solve the linear equation $\frac{dy}{dx} - \frac{2y}{x} = x + x^2$.	7	L3	2	2
c)	Solve the differential equation $(D^2 + 3D - 4)y = 12e^{2x}$	7	L3	2	2
4. a)	Solve: $(D^2 + 10D + 25)y = 0$	6	L3	3	2
b)	Solve the differential equation $\frac{dy}{dx} - e^{2x-3y} + 4x^2e^{-3y}$ by separating the variables.	7	L3	2	2
c)	Solve the differential equation $(D^3 + 4D)y = \sin 2x$ by method of variation of parameters.	7	L3	3	1

5. a) Solve: $(D^2 + D)y = x + 1$.
 b) Solve the linear differential equation $\cos^2 x \frac{dy}{dx} + y = \tan x$.
 c) Solve: $(D^2 + 36)y = 4 \cos 2x$.

6	L3	3	2
7	L3	3	2
7	L3	3	2

Unit – III

6. a) Find the Laplace transform of the function $\frac{e^{at} - e^{bt}}{t}$.
 b) Derive the formula for the Laplace transform of $\sin at$.
 c) Find the Laplace transform of the function $te^{-4t} \sin 4t$.

6	L2	4	2
7	L3	4	2
7	L3	4	1

7. a) Find i) $L\{1\}$ ii) $L\{e^{at}\}$.
 b) Find i) $L\left\{t^3 + \frac{1}{2}t^2 + 17t - 1\right\}$ ii) $L\left\{\frac{1 - e^t}{t}\right\}$.
 c) Find the Laplace transform of $\int_0^t \frac{t \cos at - \cos bt}{t} dt$.

6	L2	4	1
7	L3	4	2
7	L3	4	2

8. a) Find the inverse Laplace transform of the function $\frac{2s^2 - 6s + 5}{(s-1)(s-3)(s-2)}$ using partial fractions.
 b) State and prove convolution theorem.
 c) Find the solution of the following differential equation $x''(t) - 2x'(t) + x(t) = e^t, x(0) = 2$ and $x'(0) = 1$.

6	L2	5	2
7	L3	5	2
7	L3	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
